VLM-DM: Visual Language Models for Multitask Domain Adaptation in Driver Monitoring

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***Abstract*— Driver monitoring systems face critical challenges in modern transportation, including limited multitasking capabilities and a lack of interpretability. These limitations hinder the accurate and comprehensive assessment of driver states such as distraction, drowsiness, and emotions, which are essential to ensure road safety. This paper introduces visual language models for multitask domain adaptation in driver monitoring (VLM-DM), a novel framework that addresses these challenges by leveraging advanced visual language models for the simultaneous execution of multiple driver monitoring tasks. By employing parameter-efficient training methods such as Low-Rank Adaptation (LoRA) and integrating dynamic prompt tuning, VLM-DM achieves superior performance compared to state-of-the-art methods. Our experiments on three benchmark datasets across different driver states, demonstrating significant improvements in multitask accuracy and interpretability. This work highlights the potential of advanced multitask and multimodal architectures in developing robust, scalable, and interpretable driver monitoring systems for real-world applications.**

I. INTRODUCTION

The integration of advanced sensing technologies into vehicles has significantly improved vehicle dynamics performance and driving safety [1]–[5]. Driver monitoring systems (DMS) are becoming increasingly important due to the complexity of driver interactions with in-vehicle systems and the rise of automated driving levels, which are essential for preventing road accidents caused by driver abnormal states such as distraction [6] and drowsiness [7]. Driver monitoring can be broadly categorized into physiological signal analysis and visual monitoring. Physiological measures, such as electroencephalography (EEG), heart rate, and respiratory rate, are effective in detecting cognitive states but often require intrusive setups, limiting their practicality in real-world deployment [8]. In contrast, visual monitoring systems utilize cameras to observe behaviors and physical cues, including eye closure, yawning, and facial expressions [9]. These non-intrusive systems are more suitable for driver monitoring, thus this work focuses on visual monitoring.

Different driver states highlight unique requirements for driver monitoring. For example, distraction detection emphasizes head and upper body movements, whereas drowsiness detection focuses on close-up observations of eye closure and yawning. Since most methods primarily focus on detecting specific driver states, multiple models have to be deployed simultaneously in practical applications. Consequently, such a method complicates the system and increases the risk of failures in ambiguous situations with overlapping behaviors [9]. To address these challenges, we propose a **v**isual **l**anguage **m**odel framework specifically designed for multitask **d**river **m**onitoring (VLM-DM), aiming to monitor different driver states using a unified model. Specifically, this framework comprises three core components, that is, a vision encoder for extracting visual features, a large language model (LLM) for reasoning and decision-making, and a task-specific alignment module to bridge the visual and textual modalities. By integrating these components, VLM-DM addresses the unique requirements of driver states, such as distraction, emotion, and drowsiness. Our contributions can be summarized as follows:

- VLM-DM is designed to deal with multiple driver monitoring tasks simultaneously by aligning task-specific requirements in a cohesive framework, ensuring efficient processing and reducing redundancy.
- Through the application of Low-Rank Adaptation (LoRA) techniques, the framework achieves parameter-efficient fine-tuning (PEFT) tailored to specific datasets without compromising performance.
- A series of tests are conducted on three benchmark datasets involving distraction, emotion, and drowsiness, and the results demonstrate that our proposed method outperforms other state-of-the-art methods in terms of multitask accuracy.

The rest of this paper is structured as follows: Section II reviews related work in driver monitoring and multitask learning methods. Section III details the methodology and architecture of the proposed VLM-DM framework. Section IV presents the experimental setup and results, showing the effectiveness of VLM-DM compared to other approaches. Finally, Section V concludes the paper and outlines future work.

II. RELATED WORK

A. Driver Monitoring

In general, driver monitoring is regarded as a multi-classification problem [9], leading to the development of various supervised learning methods designed to achieve higher accuracy in recognizing specific classes. Existing studies have transitioned from traditional machine learning to

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deep learning, with the most recent advancements focusing on vision-language foundation models.

Traditional machine learning algorithms, such as support vector machines (SVMs) [10], rely heavily on manual feature extraction and cannot inherently capture information from images. In comparison, deep learning methods leverage an end-to-end training paradigm to directly utilize raw images/video clips for driver monitoring [9]. Most deep learning methods, built on CNNs and vision transformers (ViTs), are widely applied to driver monitoring tasks, including distraction, emotion, drowsiness, workload, etc. [11], [12]. For instance, a Swin Transformer-enabled weakly supervised contrastive learning method has been designed to quantitatively identify driver behaviors, showing superior recognition performance on previously unseen activities [6]. Also, various hybrid architectures that combine the strengths of different modules were developed, wherein CNN and ViT captured local details and global dependencies in images, respectively [13]. Recently, vision-language foundation models, such as CLIP and VLMs, have further enhanced semantic understanding in driver distraction detection, facilitating more intuitive user interactions [14]. Nevertheless, these models are tailored to specific driver states, making it difficult to achieve multi-state monitoring by capturing information at different scales.

B. Multitask Learning

Multitask learning (MTL) [15] has emerged as a critical paradigm in machine learning, enabling models to leverage shared representations across related tasks for improved generalization. MTL methods can be broadly categorized into hard and soft parameter sharing strategies. Hard parameter sharing allocates lower-layer parameters to multiple tasks while preserving task-specific parameters at the output layers, but this approach can cause task conflicts that are commonly described as the 'teeter-totter' phenomenon. For example, Cross-Stitch Networks [16] introduced cross-stitch units that balance shared and task-specific features through learnable weights, dynamically adjusting the contribution of features to minimize conflicts between tasks. In contrast, soft parameter sharing allows each task to maintain its parameters while encouraging similarity through regularization. Methods like Multi-Task Attention Networks (MTAN) [17] refined the backbone features by introducing task-specific attention mechanisms, ensuring that each task focuses on the most relevant parts of the feature space and enhancing task flexibility. However, this approach often comes with increased computational complexity. Recent advances have focused on improving task-specific adaptability and dynamic parameter sharing. Mixture-of-Experts (MoE) [18] assigned task-specific experts to specialize in distinct feature transformations, enabling efficient utilization of resources while maintaining adaptability. Based on this idea, TaskMoE [19] incorporated a gating mechanism to dynamically determine the contributions of multiple experts, effectively balancing shared and task-specific representations to improve scalability and performance in scenarios with diverse task requirements.

Although the above strategies provided a solid foundation for MTL in general applications, their effectiveness in DMS remains an open question. Most current approaches to driver state monitoring unify the cues needed for multiple tasks (e.g. facial features, head pose, and gaze direction) into a single model and solve all tasks simultaneously [20], thus reducing memory footprint, computational cost, and risk of overfitting. Some methods estimate the gaze direction by regressing the yaw and pitch angles of the driver's eyes with respect to the camera coordinate system. However, these approaches are highly sensitive to the physical setup, such as the camera mounting position and driver seat location, making them difficult to generalize across different vehicles. When the model is deployed on different vehicles (i.e., camera mounting location and driver seat location), it exhibits limitations. The required conditions are also very demanding.

III. Methodology

A. Framework

The VLM-DM framework is designed to achieve multitask driver monitoring, as shown in Fig. 1. Textual prompts can explicitly guide the model toward specific driver monitoring tasks, enabling generalization across domains. The core architecture consists of a vision encoder and an LLM, both of which operate collaboratively to process multimodal input and generate task-specific outputs. This modular framework enables flexible adaptation to various tasks, robust cross-modal alignment, and interpretable predictions.

The input information includes two components, i.e., visual images and textual prompts. In this work, the input image has a shape of $H \times W \times 3$, where H and W represent the height and width of the image, respectively. The images are resized during preprocessing to a consistent dimension (e.g., 224×224) compatible with the model input, then the vision encoder extracts N image patch embeddings from the input image, forming $h_V \in \mathbb{R}^{N \times 4096}$. The textual prompt T

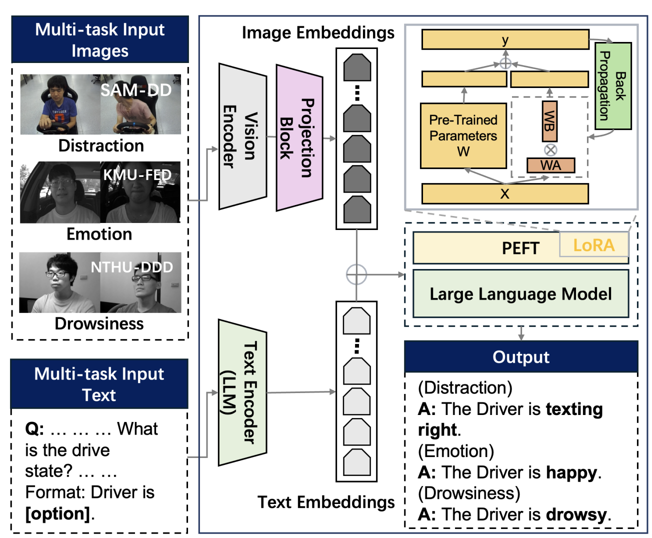

Fig. 1. Framework of the proposed VLM-DM.

is tokenized into a sequence of L tokens. Each token is embedded into a 4096-dimensional vector, resulting in the text embedding sequence $h_T \in \mathbb{R}^{L\times 4096}$. Meanwhile, These visual and textual embeddings are concatenated along the sequence (token) dimension to form a unified multimodal representation,

$$z = h_V \oplus h_T \tag{1}$$

wherein the operation $\oplus$ refers to concatenation along the sequence (token) dimension, and $z \in \mathbb{R}^{(L+N)\times 4096}$ is a unified representation that is used for both reasoning and task-specific output generation. LoRA fine-tuning injects trainable low-rank weights into the attention and feedforward layers of the LLM, the model is adapted for specific tasks while keeping the multimodal input z unchanged. The fusion of these embeddings enhances the model's ability to reason across modalities, leading to robust performance in multitask settings. Finally, the task-specific reasoning is performed by the fine-tuned LLM to yield the final output for multitask classification,

$$\hat{y} = \mathrm{LLM}(z) \tag{2}$$

wherein $\hat{y}$ denotes the text responses, that is, the class of driving behaviors, emotion, or drowsiness states. The modular design of the VLM-DM framework ensures that each component can be independently replaced, providing flexibility and scalability. For example, the vision encoder can be adapted to different visual datasets or replaced with more advanced architectures, while the LLM can be updated with improved reasoning capabilities. The use of a shared representation space simplifies the integration of multimodal inputs, enabling seamless multitask learning and reducing the complexity of adding new tasks.

B. Training Settings

The VLM-DM framework utilizes LoRA for fine-tuning, enabling parameter-efficient optimization for multitask driver monitoring without extensive computational overhead. The fine-tuning process combines a ViT as the vision encoder with the LLM Vicuna, which serves as both a text encoder and decoder for processing textual information. LoRA fine-tunes a small subset of model parameters by decomposing the weight matrices in transformer layers into low-rank forms. For a given weight matrix W, LoRA decomposes it as

$$W = W_0 + UV^T \tag{3}$$

Here, W_0 represents the original pre-trained weight matrix, while U and V are low-rank matrices learned during fine-tuning. This decomposition ensures that the total number of learnable parameters is significantly reduced compared to the full fine-tuning. The adaptation is applied to both the attention and feed-forward layers in the transformer, preserving the model's pre-trained knowledge while allowing task-specific adjustments. Accordingly, transformer-based attention mechanisms are modified as

$$\mathrm{Attention}(Q,K,V) = \mathrm{softmax}\left(\frac{Q(K+\Delta K)^T}{\sqrt{d_k}}\right)V \tag{4}$$

TABLE I

CHARACTERISTICS COMPARISON ACROSS DATASETS

Characteristics	SAM-DD [6]	KMU-FED [21]	NTHU-DDD [22]
Resolution (W × H)	1200×900	1600×1200	640×480
Image modes	RGB	Grey	Grey
Sample types	Single frame	Image sequences	Video frames
Categories	10	6	2
Sample sizes	51,175	53,217	1,106
Participants	42	12	36
Gender ratio (M / F)	34 / 8	9 / 3	6 / 0
Collection scenes	In-lab	Real driving	Simulated driving

where the query Q, key K, and value V matrices are calculated through linear projections of the hidden states, ΔK denotes the learned low-rank adaptation applied to the key matrix, and d_k is the dimensionality of the key vectors.

A single unified model is trained to recognize distraction, emotion, and drowsiness simultaneously, leveraging task-specific textual prompts during inference to enable multitask capability. Model training is carried out on the National Supercomputing Centre (NSCC) platform using two NVIDIA A100 GPUs, and mixed-precision training with BF16 is employed to accelerate computations and reduce memory usage. The training setup features a learning rate of 2×10^{-4} for LoRA parameters, managed by a cosine learning rate decay scheduler with a warmup ratio of 3.0%. The modality projector, a lightweight two-layer perceptron with GELU activation, is fine-tuned with a learning rate of 2×10^{-5}.

IV. EXPERIMENTAL SETTINGS

A. Dataset Features

This work employs three datasets across different driver states, i.e., distraction, emotion, and drowsiness, to validate the models' multitask capability. The dataset features are summarized in Table I, with corresponding descriptions provided. We follow the official data splits provided by each dataset for training and testing to maintain consistency with prior works.

1) SAM-DD [6]: The Singapore AutoMan@NTU Distracted Driving dataset includes RGB and depth videos from 42 participants (34 males, 8 females) exhibiting ten driving behaviors. These behaviors are captured at a resolution of 1200×900 pixels using two synchronized cameras. Our work focuses exclusively on the RGB data, leveraging this dataset to address diverse distracted behaviors in multitask settings.

2) KMU-FED [21]: The Keimyung University Facial Expression of Drivers dataset is designed to recognize facial expressions in real driving environments. It includes 55 image sequences from 12 subjects, featuring six emotional states. The data was captured using NIR cameras mounted on the dashboard or steering wheel, incorporating variations in lighting and occlusions (e.g., hair or sunglasses). This dataset highlights the challenges of emotion recognition in dynamic and unconstrained driving conditions.

3) NTHU-DDD [22]: The Driver Drowsiness Detection Dataset from National Tsing Hua University's Computer Vision Lab focuses on identifying driver drowsiness. It features recordings of 36 subjects in simulated driving scenarios, showcasing two levels of drowsiness, i.e., non-drowsy and drowsy. The dataset comprises video sequences under both daytime and nighttime conditions, provided at resolutions up to 640×480, and frame rates ranging from 15 to 30 frames per second. This dataset is crucial for understanding driver drowsiness in controlled environments.

B. Baselines

This section describes the baseline approaches used in our experiments. We evaluate different combinations of widely used backbones and multitask learning frameworks to provide a comprehensive comparison with our proposed method.

Backbones. The backbone architectures were carefully selected to cover a wide spectrum of network design paradigms, which can be classified into two groups, i.e., CNN-based and transformer-based. CNN-based backbones, including ResNet50, MobileNet, and EfficientNet, are widely recognized for their extensive use in image-based tasks and have historically served as baselines for conventional approaches. ViT, a representative Transformer-based method, was selected to represent state-of-the-art advancements in visual tasks. Its self-attention mechanism is particularly effective for capturing long-range dependencies and complex patterns, making them well-suited for multitask learning settings. These backbones were chosen for their diversity, practicality, and suitability for multitask learning. Other well-known backbones like Swin Transformer and DenseNet were excluded due to higher computational costs and limited benchmarks in multitask scenarios.

All backbones were initialized with weights pre-trained on ImageNet, which provides a strong starting point by leveraging generic visual features. In our experiments, these backbones were further trained on our multitask datasets without freezing any layers. This approach allows the backbones to adapt to task-specific nuances while retaining the benefits of pre-trained features. The output from the backbone is fed into three task-specific classification heads, each comprising a fully connected layer.

MTL Frameworks. To comprehensively evaluate multitask learning capabilities, we selected four distinct frameworks, i.e., CrossStitch, MTAN, MoE, and TaskMoE. Each framework represents a different approach to sharing and specialization, offering insights into the trade-offs between parameter sharing and task-specific learning.

1) CrossStitch [16]. This framework employs cross-stitch units to balance shared and task-specific features. Learnable cross-stitch weights dynamically adjust the contribution of features from the shared backbone and task-specific streams. The shared output features are used for further task-specific classification.

2) MTAN [17]. This framework introduces task-specific attention mechanisms to refine backbone features, and each task only focuses on relevant parts of the feature space.

3) MoE [18]. This work assigns task-specific experts to specialize in distinct feature transformations. Each task uses a specific expert without additional gating mechanisms.

4) TaskMoE [19]. An enhanced MoE framework with a gating mechanism that learns the contributions of each expert dynamically. The final output combines expert predictions using the learned weights.

V. Results and Discussions

In this section, we conduct a detailed performance analysis of the proposed VLM-DM framework. Multitask models' performance is evaluated by comparing various backbones and frameworks across three datasets covering different driver states. Also, the confusion matrices and attention maps are provided to demonstrate the model's effectiveness further.

Backbone Comparison. Table II highlights the performance of different backbones, including ViT (Ours), ResNet, and MobileNet. The results demonstrate that ViT consistently outperforms other models in both accuracy and stability

TABLE II
BACKBONE COMPARISON ON ACCURACY AND F1 SCORE ACROSS DATASETS

Model/Method	SAM-DD		FED		NTHU-DDD		Overall	
	Accuracy	F1 Score	Accuracy	F1 Score	Accuracy	F1 Score	Accuracy	F1 Score
ViT (Ours)	0.920	0.918	0.979	0.979	0.993	0.993	0.964	0.963
ResNet	0.907	0.905	0.988	0.988	0.994	0.994	0.963(↓0.1%)	0.963(↓0.0%)
EfficientNet	0.904	0.900	0.985	0.985	0.991	0.991	0.960(↓0.4%)	0.959(↓0.4%)
MobileNet	0.889	0.888	0.988	0.988	0.996	0.996	0.958(↓0.6%)	0.957(↓0.6%)

TABLE III
COMPARISON OF VLM-DM WITH VIT BACKBONE ACROSS DIFFERENT MULTITASK FRAMEWORKS

Model/Method	SAM-DD		FED		NTHU-DDD		Overall	
	Accuracy	F1 Score	Accuracy	F1 Score	Accuracy	F1 Score	Accuracy	F1 Score
VLM-DM (Ours)	**0.973**	**0.987**	**0.976**	**0.988**	**0.979**	**0.979**	**0.976**	**0.985**
CrossStitch	0.922	0.921	0.943	0.942	0.993	0.993	0.953(↓2.3%)	0.952(↓3.3%)
MTAN	0.854	0.860	0.994	0.994	0.994	0.994	0.947(↓2.9%)	0.949(↓3.6%)
MoE	0.889	0.884	0.982	0.982	0.930	0.930	0.934(↓4.2%)	0.932(↓5.3%)
TaskMoE	0.897	0.896	0.163	0.046	0.542	0.380	0.534(↓44.2%)	0.441(↓55.4%)

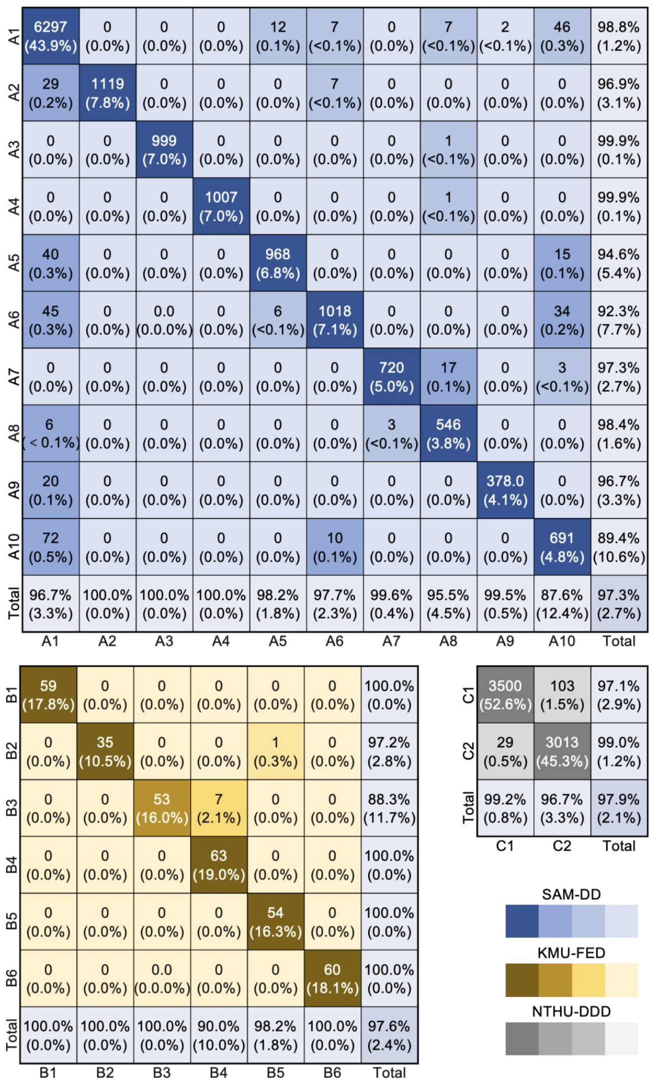

Fig. 2. Confusion matrices for SAM-DD, NTHU-DDD, and KMU-FED datasets. The matrices display the classification results for multitask driver behavior detection. **A1–A10** denote different driver behaviors for the SAM-DD dataset, corresponding to Normal Driving, Drinking, Phoning Left, Phoning Right, Texting Left, Texting Right, Touching Hairs & Makeup, Adjusting Glasses, Reaching Behind, Dropping, respectively. **B1–B6** denote different emotions in the KMU-FED dataset, corresponding to Angry, Disgust, Fear, Happy, Sad, and Surprise, respectively. **C1–C2** denote drowsiness levels in the NTHU-DDD dataset, corresponding to Non-Drowsy and Drowsy, respectively.

across all datasets. For instance, on the SAM-DD dataset, ViT achieves an accuracy of 92.0%, surpassing MobileNet by 3.1%. On the NTHU-DDD dataset, ViT not only achieves competitive accuracy of 99.3% but also demonstrates superior stability with a standard deviation of ±0.031 in both accuracy and F1 score. These results demonstrate that the self-attention mechanism effectively captures intricate relationships both within and across modalities, providing a significant advantage in integrating visual inputs with natural language representations. Also, ViT's tokenization strategy aligns well with language models, making it ideal for cross-modal tasks.

Framework Comparison. Based on the same backbone (ViT), Table III evaluates the performance of VLM-DM against four multitask frameworks. It is observed that TaskMoE exhibits significantly low accuracy due to its gating mechanism's sensitivity to initialization and lack of robust regularization, such as entropy constraints. This leads to a 'collapse', where tasks over-rely on a single expert, preventing the model from effectively utilizing multiple experts. CrossStitch and MTAN perform worse than VLM-DM due to their inability to adapt to varying task relationships dynamically. These frameworks are prone to negative transfer, where conflicting task objectives produce suboptimal shared feature representations, especially in diverse datasets like SAM-DD and FED. VLM-DM leverages ViT's cross-modal representation capabilities, which are essential for tasks like emotion recognition and drowsiness detection, which demand fine-grained visual understanding. This is particularly relevant to KMU-FED, where nuanced facial expression distinctions require robust feature extraction.

Fig. 3. Attention maps comparison before and after fine-tuning for multi-task driver monitoring using attention maps. (a) "Touching" and "Phoning Right" behaviors in the SAM-DD dataset. (b) "Angry" and "Disgust" emotions in the KMU-FED dataset. (c) "Drowsy" and "Non-Drowsy" in the NTHU-DDD dataset.

Confusion Matrix Analysis. The confusion matrices of three datasets provide detailed insights into the classification performance of VLM-DM, as shown in Fig. 2. The results demonstrate VLM-DM's consistent recognition performance across diverse monitoring tasks, highlighting its generalizability and robustness in multitask settings. Nevertheless, the model remains prone to confusing certain behaviors. In the SAM-DD dataset, *Reaching Behind* (A9) and *Dropping* (A10) exhibit relatively higher confusion rates with categories like *Normal Driving* (A1) and *Texting Right* (A6). This phenomenon suggests that the model faces challenges in distinguishing actions with overlapping visual features or limited motion dynamics. Similarly, the subtle expression *Fear* (B3) is frequently misclassified in the KMU-FED dataset, potentially due to the small inter-class variation and limited sample size. Incorporating temporal information to disambiguate sequential actions is a potential approach to addressing misclassification in subtly different categories. Also, the inclusion of additional training samples or synthetic data augmentation for underrepresented categories could further enhance performance.

Qualitative Analysis. To intuitively understand the effectiveness of LoRA fine-tuning, a VLM-Visualizer approach is employed to visualize the attention maps further. As shown in Fig. 3, our model demonstrates the ability to focus on different regions in the three datasets, demonstrating its ability to capture visual information at various scales. After LoRA fine-tuning, the model targets the upper body, face, and eyes across various monitoring tasks. This highlights the effectiveness of the model in focusing on task-relevant regions based on the nature of the task. For example, the model correctly focuses on the driver's hands when detecting behaviors such as "Touching Hairs & Makeup" in distraction detection, as shown in Fig. 3(a). In contrast, the model focuses on the driver's eyes for drowsiness detection, as shown in Fig. 3(c). On the whole, LoRA fine-tuning assists the model in focusing on relevant visual regions, thereby reinforcing VLM models' performance on specific tasks.

VI. Conclusion

In this paper, we proposed VLM-DM, a novel VLM-based framework for multitask domain adaptation in driver monitoring. By integrating advanced techniques such as LoRA, VLM-DM effectively addresses the challenges of traditional systems, demonstrating superior accuracy, robustness, and interpretability across tasks like distraction detection, emotion recognition, and drowsiness detection. Experimental results show that VLM-DM not only outperforms state-of-the-art methods but also unifies diverse driver state monitoring tasks under a single, scalable framework, reducing computational cost and complexity while enhancing adaptability. This work provides a flexible solution for improving driver safety and offers potential extensions to broader applications within autonomous driving, such as shared control and takeover strategies [23], [24]. In the future, computational efficiency could be optimized by lightweight transformer variants or hybrid architectures combining CNNs and ViTs. Additionally, the model generalization across varying conditions could be improved by domain adaptation techniques.

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